

ABDUL MUNIM BAIG

Software Engineer, Backend Systems & Applied AI/ML

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PROFESSIONAL SUMMARY

Computer Science graduate with hands-on experience spanning backend systems, database internals, and applied machine learning. Comfortable writing C and PostgreSQL extensions at the systems level and building RAG pipelines and LLM-based applications at the application layer. Experienced in Linux/Ubuntu environments, collaborating asynchronously with distributed engineering teams.

EDUCATION

Bachelor of Science in Computer Science

2022 to 2026

FAST National University of Computer and Emerging Sciences, Islamabad

Relevant Coursework: Operating Systems (Linux, Bash/shell scripting), Computer Networks, Network Security (cryptography fundamentals), Database Systems, Deep Learning, Generative AI, DevOps Engineering (Docker, Kubernetes basics, CI/CD, Git)

EXPERIENCE

Software Engineer Intern, INZENT (Remote)

Mar 2026 to Aug 2026

- Worked fully remote on PostgreSQL-based database tools in a Linux/Ubuntu environment, writing and debugging C-based database packages and extensions at the system level while collaborating asynchronously with a distributed engineering team.
- Contributed to a PostgreSQL extension project intended for future open-source release, currently in private, internally-restricted development.
- Researched PostgreSQL internals to identify performance bottlenecks and improve reliability of existing database packages.

IT Department Intern, Pakistan Tobacco Company

Jun 2025 to Jul 2025

- Designed and deployed a Leave Management System using Microsoft Power Platform, automating approval workflows and reducing manual steps in internal HR processes.

Machine Learning Engineer Intern, Genesys Research Lab

Jun 2024 to Aug 2024

- Preprocessed and analyzed large-scale datasets used for training machine learning models.
- Built backend data pipelines with Node.js and MongoDB to support model training workflows.

PROJECTS

AIVA: AI Virtual Assistant (Final Year Project) (WordPress, Python, Flask, Docker, Google Cloud Run, Groq)

- Built an AI-powered commerce engine for WordPress automating customer engagement, order handling, and payment verification across social platforms, with a Flask REST API exposing the chatbot logic and NLP-driven intent detection.
- Architected a dual-LLM setup (Groq Llama 3.3 + a local 7B model) balancing fast reasoning with conversational depth, and used a vector database (Sentence-Transformers) for semantic product search and RAG-based query handling.
- Containerized the service with Docker and deployed it on Google Cloud Run for scalable, on-demand hosting.

Hybrid RAG: Retrieval-Augmented QA with Hybrid Search and Reranking (Python)

- Built a RAG pipeline combining dense and sparse (BM25) retrieval with a cross-encoder reranker, and a custom evaluation set to measure retrieval precision and answer faithfulness.

Multimodal RAG System for Document Analysis (Python, FAISS, CLIP, LLaMA 3)

- Built an end-to-end RAG pipeline for complex PDFs, combining text and image embeddings for retrieval.

Resilient Agent: Multi-Step AI Agent with Tool Use and Recovery (Python)

- Built an autonomous AI agent capable of multi-step reasoning, external tool invocation, persistent memory, and automatic retry/recovery when an intermediate step fails.

TECHNICAL SKILLS

Languages: Python, C, C++, Java, JavaScript, SQL

AI & Machine Learning: LLM Engineering, RAG, NLP, Deep Learning, Generative AI, Information Retrieval, Semantic Search, Embeddings, TensorFlow, PyTorch, NumPy, Sentence-Transformers, FAISS

Systems & Tools: Linux (Ubuntu), PostgreSQL internals, Docker, Google Cloud Run, Git/GitHub

Databases: PostgreSQL, MySQL, MongoDB, Vector Databases (ChromaDB, FAISS)

Web & Frameworks: Flask (REST APIs), React, Spring Boot, Node.js

Security & Networking: Network security, cryptography, computer networks